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Catalytic hydrogenation of CO_2 from 600 MW supercritical coal power plant to produce methanol: A techno-economic analysis

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ABSTRACT

Electricity and water from renewable hydropower plant are used as input for electrolysis unit to generate hydrogen, while CO_2 is captured from 600 MW supercritical coal power plant using post-combustion chemical solvent based technology. The captured CO_2 and H_2 generated through electrolysis are used to synthesize methanol through catalytic thermo-chemical reaction. The methanol synthesis plant is designed, modeled and simulated using commercial software Aspen Plus®. The reactor is analyzed for two widely adopted kinetic models known as Graaf model and Vanden-Bossche (VB) model to predict the methanol yield and CO_2 conversion. The results show that the methanol reactor based on Graaf kinetic model produced 0.66 tonne of methanol per tonne of CO_2 utilized which is higher than that of the VB kinetic model where 0.6 tonne of methanol is produced per tonne of CO_2 utilized. The economic analysis reveals that 1.2 billion USD annually is required at the present cost of both H_2 production and CO_2 abatement to utilize continuous emission of 3.2 million tonne of CO_2 annually from 600 MW supercritical coal power unit to synthesize methanol. However, sensitivity analysis indicates that methanol production becomes feasible by adopting anyone of the route such as by increasing methanol production rate, by reducing levelised cost of hydrogen production, by reducing CO_2 mitigation cost or by increasing the current market selling price of methanol and oxygen.

Introduction

The contribution of carbon dioxide (CO_2) to the global warming that leads to various detrimental impacts such as rise in sea level, melting of glaciers and rise in average global temperature makes it imperative to deal stringently with the direct emission of CO_2 to atmosphere. To mitigate CO_2 there have been taken several global initiatives such as United Nations Framework Convention on Climate (UNFCCC), 2030 framework, COP21 agreement and 2050 roadmap. However due to growing energy demand globally will keep the trend of

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installing fossil fuel based power plants [1]. In this scenario one of the strategies called Carbon Capture and Storage (CCS) is commonly suggested in which initially the CO_2 is captured from the source (power plants, petrochemicals, cement industry, paper mill, and steel mills etc.) and then this captured CO_2 is sequestered to deep saline water [2]. The other available option is to convert captured CO_2 into useful chemical or fuel instead of storing to deep sea water.

Carbon Dioxide Utilization (CDU) has recently gained much attention as an alternate option to convert captured CO_2 to various hydrocarbons such as methanol (MeOH), methane (CH_4), formic acid, carbon monoxide (CO), and dimethyl ether (DME) through catalytic hydrogenation [3–6]. Among various products formation options resulting from catalytic CO_2 hydrogenation, methanol synthesis through CDU is an attractive fuel, alternate to gasoline or can be used directly in fuel cell and as a key chemical feedstock for various chemical production [7]. Conventionally, methanol is either produced through the CAMERE (carbon dioxide hydrogenation to methanol via reverse water gas shift) process or through direct methanol synthesis [8]. The later method is used for the conversion of captured CO_2 to methanol takes place through direct methanol synthesis. Three distinct opinions exist about the source of carbon atom in methanol. First research group derived the kinetic model based on the notion that CO is the main source of carbon in methanol synthesis [9–11]. The second group suggested that CO_2 is the contributor of carbon [12–14], whereas the third group predicted that both CO_2 and CO contribute simultaneously in the formation of methanol [15–19]. Among various kinetic models, the model of Graaf [15] and Vanden Bussche (VB) model [12] are considered reliable and thus commonly adopted.

Various aspects of methanol production through catalytic CO_2 hydrogenation are explored in the literature. Van-Dal and Bouallou [20] adopted Aspen Plus as modeling tool to present a simulated model of methanol plant based on the VB kinetic model and predicted that 1.6 t of CO_2 is utilized per tonne of methanol produced. However, the results are presented without economic assessment and without comparing with other kinetic models. Kiss et al. [21] proposed flowsheet of methanol production plant through CO_2 hydrogenation based on Graaf kinetic model, Whereas Jia et al. [22] and Murciano et al. [23] analyzed possible hydrocarbon formation through CO_2 hydrogenation by using Gibbs free energy minimization technique without considering reaction kinetic model. Both of the articles lack economic assessment and comparison with other kinetic model. Anicic et al. [8] presented techno-economic comparison between direct methanol synthesis and the CAMERE process and concluded that the former method is more energetic efficient and economical. Another similar techno-economic study by Fortes et al. [24] based on direct methanol synthesis using CDU and VB kinetic model, the authors predicted that CO_2 conversion in the reactor can be achieved up to 22% and CO_2 abatement was 1.46 tonne per tonne of methanol produced. However, study just considered CO_2 as carbon source in methanol production without considering CO production. They concluded that based on the current price of hydrogen in Europe, methanol synthesis through catalytic CO_2 hydrogenation is not economically feasible. Likewise, to make catalytic CO_2 hydrogenation economically feasible, Sayah et al. [25] predicted that the current price of H_2 production and cost of CO_2 capture must be reduce at least by a factor of 10 and a factor of 2, respectively.

The notion of utilizing captured CO_2 with renewable energy resources such as solar, wind and hydropower to produce methanol seems to be an attractive option to control carbon footprints and recycle CO_2, however, cost of methanol production is the crucial issue. Based on the Italian scenario, Rivarolo et al. [26] presented economic feasibility of methanol production through hydrogenation of CO_2 obtained from two different sources, the one captured CO_2 from bio-methane plant and the other directly purchased from the market. Their results indicated that methanol produced from market purchased CO_2 is not profitable considering current selling price of methanol. However, with methanol selling price of 500 € per ton or 600 € per ton coupled with either solar or wind renewable energy source, the methanol production can become economically feasible. Likewise, Bellotti et al. [27] performed economic feasibility of methanol production using captured CO_2 from fossil fuels based on the German scenario. They predicted that three important factors such as oxygen production as by-product, cost of electrolyzer and current selling price of methanol are the crucial parameters to be seriously addressed for an economically feasible methanol production. By exploring the installed methanol production plant, the first methanol generation plant started production in 2012 in Iceland that utilized captured CO_2 from a geothermal power station and electricity from hydro and geothermal power plant [28]. In 2015, the plant increased its annual methanol production capacity to 1.3 million liters by utilizing 5.5 thousand tons of CO_2. Rivarolo et al. [29] analyzed a similar concept of integrating one of the largest hydropower plant Itaipi with installed capacity of 14 GW owned by Brazil and Paraguay, with biomass gasification for co-production of hydro-methane and methanol.

The predicted methanol yield and CO_2 conversion in the catalytic reactor mainly depends on the type of catalyst used in the reactor and the operating conditions. The predicted methanol yield and CO_2 conversion based on the Graaf kinetic model can be much different in value than that obtained from the VB model and hence subsequent production cost analysis can produce different results. In most of the work presented in literature, authors adopted either the Gibbs free energy minimization technique without considering catalyst that provide rough estimation of methanol yield or they considered VB kinetic model alone or Graaf model. Moreover, studies related to economic feasibility of methanol synthesis from CO_2 are presented without considering one of the many factors which include the optimal operating conditions, or without considering the actual integration with flue gas source and H_2 production source, and detailed cost factors in the related input system. Therefore, there is a need to analyze utilization of captured CO_2 from the flue gases of coal power plant at large scale for methanol production under optimal operating conditions based on two distinct opinions of carbon source in the methanol, and then based on the methanol produced a detailed cost calculation methodology is required for comprehensive understanding of the integrated system.

In this study, a novel idea is presented by integrating a renewable hydropower source, coal power plant and proton exchange membrane (PEM) electrolysis with the methanol production plant. Water and electricity input for electrolysis system is obtained from hydropower plant to generate H_2, whereas CO_2 is captured from 660 MW supercritical coal power plant using chemical solvent based absorption-desorption method. In this paper, the idea of using electricity and water from one of the largest hydro power plant of China is adopted due to the fact that electricity from large-scale hydropower plant is the cheapest compared to other renewable (solar and wind) sources [30]. Also, water from run of the dam is in abundance and free to use in electrolyzer cell to generate hydrogen. Moreover, the methanol production may be intermittent because the electricity from solar or wind power systems is not available continuously throughout the day. The methanol synthesis plant is designed and simulated using commercial tool Aspen Plus® V8.6, after optimizing operating conditions, catalyst loading and reactor size, the maximum methanol yield and CO_2 conversion is predicted based on the Graaf and VB kinetic model. After obtaining optimal methanol yield from the methanol plant, the economic feasibility of methanol production by catalytic CO_2 hydrogenation is analyzed for the complete integrated system. Also, sensitivity analysis is performed to analyze the impact of crucial factors such as production rate of methanol, cost of hydrogen generation, cost of variable expenses, market sale price of methanol and oxygen, on the total worth of products produced and on total cost of production. The findings of this study provide not only the technical procedure for the utilization of CO_2 to synthesis methanol at large scale but also gives comprehensive cost estimation and future prospects on how to make this route economically feasible.

Process description

The conceptual block diagram of the complete system is shown in Fig. 1. The methanol synthesis is carried out in catalytic reactor through thermo-chemical reaction of H_2 with CO_2. The hydrogen is produced through water electrolysis using PEM electrolysis technique. The electricity and water for electrolysis is obtained through large hydro power plant. The second component of feed stock for methanol synthesis is CO_2 which is obtained from flue gases of coal power plant. The CO_2 from flue gases is captured using the conventional chemical solvent based absorption-desorption method. The captured CO_2 is purified and compressed to supply for the methanol synthesis plant. In methanol plant, CO_2 reacts with hydrogen in the catalytic reactor to produce methanol. All the sections in the system are described as followed:

Hydro power plant

Currently, hydro power is the only clean energy that is economically feasible at large scale and represents 20% of the electricity generated worldwide [31]. China has installed about 145 GW of hydro power plants which contributes 6.7% of the total electricity generated with world's largest hydro power plant known as "The Three Gorges" having total capacity of 22.4 GW [32]. According to Intergovernmental Panel on climate Change (IPCC) report, the hydro power plants are either constructed as run-of-river type which utilizes water flow of the river, or storage type hydro power plant. The storage type hydropower plants store large capacity of water in dam and are usually of large capacity compared to run-of-river type hydropower plant. Based on the electricity generating capacity, hydro power plants are characterized as Pico ($\leq$ 5kW), Micro(5 – 100 kW), Mini(100kW – 1MW), Small (1 – 20 MW), Medium(20 – 100 MW), and Large($\geq$ 100 MW) hydro power plant [33]. The cost of electricity generation through hydropower varies depending on various factors such as the installed capacity of the hydropower plant, discount rate and the cost of land in the specific region. The estimated levelised cost of electricity generation from hydropower plant per kWh for Mekong region is USD 0.04–0.12 [34], for Indian region the cost is USD 0.196/kWh [35], and for USA the cost is USD 0.012/kWh, and the global cost presented is USD 0.02–0.27/kWh [36]. The estimated levelised cost of electricity generation in hydropower plant for small installed capacity of less than 10 MW is USD 0.083/kWh, and levelised cost for installed capacity of 15 MW is USD 0.06/kWh. Whereas in case of hydropower plants of capacity equal or greater than 1200 MW, the levelised cost of electricity generation is 0.016 USD/kWh [37]. Likewise, by changing the discount rate the levelised cost of electricity generation changes such as the cost range is USD (0.011–0.078) per kWh for 3% discount rate, USD (0.018–0.11) per kWh for 7% discount rate and USD (0.024–0.15) per kWh for 10% discount rate [38].

In current study, it has been assumed that the electricity and water are obtained from large scale hydropower plant such as "The Three Gorges of China" hydropower plant in which levelised cost of electricity generation is USD 0.016/kWh [32]. Based on the electricity and water input from this large scale hydropower source, the cost of H_2 generation in PEM electrolysis is computed.

CO_2 capture from 600 MW supercritical coal power plant

CO_2 capture methods are classified as absorption, desorption, membrane based, cryogenic separation and hybrid technique [39]. Among these techniques, chemical solvent based CO_2 capture is most commonly used method because this is a mature technology and can be retrofitted easily with the existing plant being the most reliable technology for post-combustion CO_2 capture from conventional coal power plants. For chemical solvent based post-combustion capture, variety of solvent options are available that are developed for CO_2 capture. It includes primary amines, secondary amines, tertiary amines, sterically hinder amines and inorganic compounds [40]. Among these solvents, monoethanolamine (MEA) is a benchmark solvent in chemical absorption and mostly adopted chemical solvent for large scale CO_2 capture because it is reliable and being used for long periods [41].

In this study, CO_2 is assumed to be captured from 600 MW through the post-combustion capture using MEA solvent. The technique involves absorption of CO_2 in absorber column and regeneration of solvent in the stripper column as described comprehensively by various researchers [42–44]. The cost calculation method of CO_2 mitigation based on 600 MW supercritical coal power plant is adopted from reference [45] in

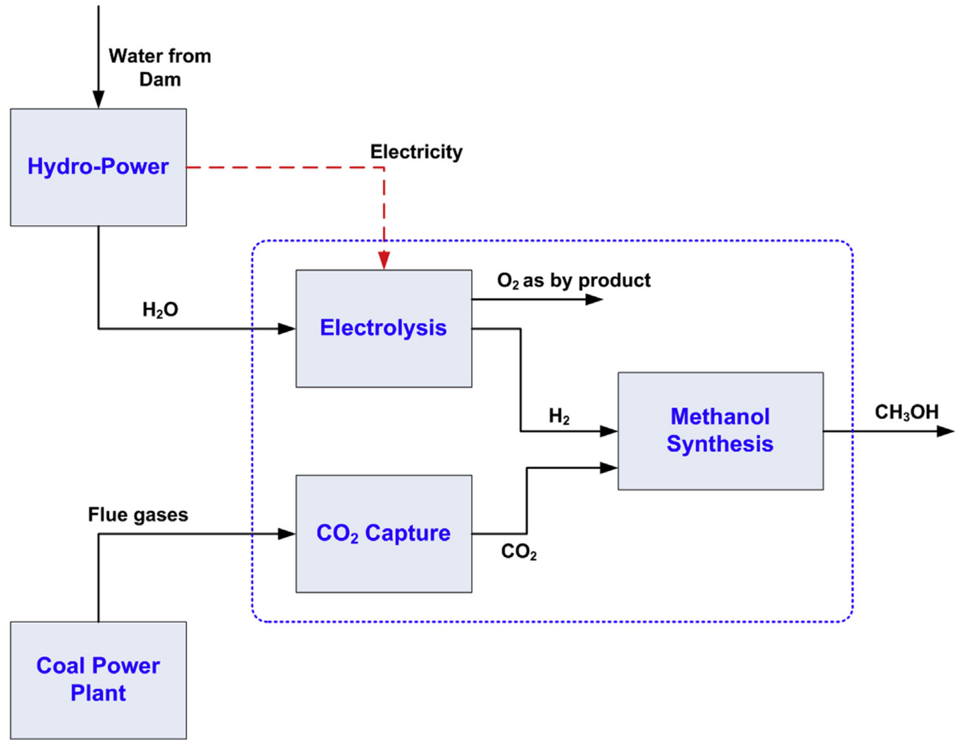

Fig. 1 – Conceptual block diagram of the integrated system for methanol sysnthesis through utilization of captured CO_2.

which comparison of CO_2 mitigation cost using various CO_2 capture techniques has been presented.

Hydrogen generation by PEM electrolysis

In electrolysis water splits into H_2 and O_2 when direct current is passed through it with the aid of electrolyte. H_2 is collected at cathode and O_2 at anode. Water electrolysis is usually classified into two types such as Alkaline Water Electrolysis (AWE) and Proton exchange membrane (PEM) Electrolysis. AWE is relatively mature technology in which liquid electrolyte is used, whereas in PEM the solid polymer based electrolyte is used. A typical electrolysis unit consists of stack, power electronics, control system, gas separation, drying unit and water circulation [46]. At standard conditions (25 °C, 1 atm), 8.9 L of water and 39.4 kW h of electricity is required to produce 1 kg of H_2 [47]. However, ideal conditions are hardly achieved due to heat generation and material limitations, therefore performance is evaluated in terms of Faradaic and energy efficiency. For 1 kg of H_2 production in the electrolysis, 8 kg of O_2 is generated at the anode as by product. Several companies have developed small and large scale electrolyzer which are commercially available, for example PEM electrolyzer is developed by Giner-USA, Siemens (Germany), Proton OnSite-USA, Hydrogenics-Canada, EREVA H2Gen (France) and ITM Power (UK). Whereas, the prominent manufacturers producing AWE electrolyzers at large scale include IHT (Switzerland), ELB (Germany), Verde (USA), Suzhou Jungli (China), NEL Hydrogen (Norway), Mcphy (France), PERIC (China), and Teledyne(USA) [48,49].

The selection of appropriate electrolyzer system to generate H_2 for the integrated system to produce methanol is a crucial task. The important factors such as current, power density, energy and Faradaic efficiency, durability of stack, constant voltage operation over the time period, capacity of plant and cost of electrolyzer, are usually considered during the electrolyzer selection. Although AWE is much matured technology than the PEM systems, but the current density of AWE system is in the range of 200–600 mA/cm^2 with low energy efficiency. On the other hand, recently developed PEM systems operate in the range of 1000–2000 mA/cm^2 [50] with high energy density are more durable and produces constant voltage over a longer period of time [51]. Moreover, the product stream such as H_2 is generated at high pressure of about 30 bar, hence save the compressing cost of H_2 [52]. However, cost of PEM electrolyzer and capacity for large scale operation are the two major hindrances for its commercial utilization at large scale.

The growing trend in green hydrogen fueling in the transport system and utilizing CO_2 with hydrogen to produce methanol shows that the demand for electrolyzer system has increased. Many investors and manufacturers are engaged in developing electrolysis system at low price and with large-scale capacity. AWE being the old and mature technology has been installed at several places at large scale. For example, NEL hydrogen has installed 168 AWE units with total capacity of 30,000 Nm^3 H_2/hr (2696 kg/hr) and with total power consumption of 135 MW [53], whereas IHT is operating AWE unit (S-556 model) of capacity 21,000 Nm^3 H_2/hr since 35 years [54]. Likewise, Aswan Electrolyzer is also producing 21,000 Nm^3 H_2/hr capacity in Zimbabwe and 5200 Nm^3/hr in Peru [55]. In contrast, PEM electrolyzer system is relatively new and has been developed at less production capacity compared to AWE. Recently several manufacturers have developed PEM electrolyzer at megawatt (MW) scale. For example Siemens successfully commissioned 6 MW PEM electrolyzer at Mainz Energy Farm Germany, the unit has three stacks with each having capacity of 2.1 MW [56]. Giner (USA) has already developed PEM electrolyzer with capacity of 400 Nm^3 H_2/hr (equal to 2 MW) and announced to launch PEM electrolyzer stack with capacity of 1100 Nm^3 H_2/hr (approximately equal to 6 MW_{el}) [57]. Likewise, NEL hydrogen reported that they have designed world's largest PEM electrolyzer based hydrogen plant of capacity 400 MW in collaboration with local industry [58].

In the current study, to adjust the stoichiometric ratio of 90% captured CO_2 with the H_2, a large scale electrolysis plant is required that can generate 51.6 tonne/hr. of H_2. With recent advancement and large scale availability of PEM electrolyzer commercially, the current analysis is based on the PEM electrolyzer whose single unit can generate 2400 kg/day, with total units of 506 and total system power rating of 2580 MW with 98% equipment availability as mentioned in Table 2.

Methanol synthesis

Conventionally, methanol is produced from syngas (mixture of CO and H_2) which can be obtained either from gasification of coal or through steam reforming [59]. However, the idea of catalytic CO_2 hydrogenation to produce methanol is new. In catalytic CO_2 hydrogenation, two outcomes are possible through reaction mechanism. One is the methanol formation and the other is reverse water gas shift reaction (WGSR) which attains equilibrium rapidly. Reaction rate of methanol synthesis is slower than the reverse WGSR which attains equilibrium faster. The methanol reactor has been analyzed using two types of kinetic reaction models, i.e. Graaf model and VB model.

The Graaf kinetic model is based on the finding that both CO_2 and CO are the source of carbon in methanol synthesis and following three reactions are considered for the model:

$$CO + 2H_2 \rightleftharpoons CH_3OH \quad \text{(Methanol from CO)} \tag{1}$$

$$CO_2 + H_2 \rightleftharpoons CO + H_2O \quad \text{(Reverse WGSR)} \tag{2}$$

$$CO_2 + 3H_2 \rightleftharpoons CH_3OH + H_2O \quad \text{(Methanol from } CO_2) \tag{3}$$

VB model considers CO_2 as the source of carbon in methanol synthesis and is based on the following reactions:

$$CO_2 + 3H_2 \rightleftharpoons CH_3OH + H_2O \quad \text{(Methanol from } CO_2) \tag{4}$$

$$CO_2 + H_2 \rightleftharpoons CO + H_2O \quad \text{(Reverse WGSR)} \tag{5}$$

In this study, both Graaf and VB models are used to compare the performance of the methanol production plant.

Modeling and simulation procedure

The model of methanol synthesis plant is developed using commercial tool Aspen Plus® V8.6. The flow sheet of complete methanol plant is shown in Fig. 2. The stoichiometric hydrogen stream (10) at mass flow rate of 52 tonne/hr., 91 °C

Table 1 – H_2 production cost.

Technical parameters	
Production equipment availability (%)	98%
Plant design capacity (kg of H_2/day)	123,840,0
Plant required output (kg/day)	121,363,2
Single unit size (kg/day)	2400
Total units required	506
Power rating of each unit (MW)	5.1
Plant life	40
System Energy (MW)	2580
Salvage value (% of total capital investment)	10%
Tax Rate (%)	6%
Direct capital costs	
Capital cost ($/kW)	587.5
Installation factor	1.12
Electrolyzer Stacks	47%
Bill of Plant Total	53%
Further division	
Hydrogen Gas Management System-Cathode system side	9%
Oxygen Gas Management System-Anode system side	3%
Water Reactant Delivery Management System	5%
Thermal Management System	5%
Power Electronics	21%
Controls & Sensors	2%
Mechanical Balance of Plant plumbing/ copper cabling/Dryer valves …	5%
Item Breakdown- Other	1%
Item Breakdown-Assembly Labor	2%
Total Direct Capital Cost ($)	**$1,538,125,762**
Indirect Capital Costs	
Cost of Engineering Design (% of direct capital cost)	8%
Project contingency (% of direct capital cost)	15%
Construction costs/site preparation (% of direct capital cost)	2%
Land cost ($/acre)	$50,000
Land required (acres)	50
Land Cost ($)	$2,500,000
Total Capital Cost ($)	**$1,965,035,762**
Operation and Maintenance Cost-Fixed	
Cost of Annual Planned maintenance (% of direct capital cost)	3%
Working staff on plant	50
Labor rate, including overheads ($/man-hr)	$50
Labor cost, $/year	
Property tax and insurance rate (% of total capital investment per year)	2%
Total Fixed Operating Cost	**$95,429,915**
Operation and Maintenance Cost-Variable	
Electricity consumption (kWh/kg H_2)	50
Electricity price ($/kWh)	0.02
Cost of electricity ($)	442,975,680
Process water consumption (gal/kg H_2)	4.76
Price of process water ($/gal)	0.001807666
Cost of process water	
Total Variable Operating Cost ($/year)	**$443,128,001**
Replacement Schedule	
Replacement time for stacks and other crucial components (years)	7
Replacement cost (% of installed capital cost)	15%
Levelized Cost of Hydrogen Production ($/kg H_2)	3.9

Table 2 – CO_2 mitigation cost analysis.

Technical parameters	
Power Plant type	600 supercritical coal power plant
Net plant efficiency (%)	27.9
Efficiency penalty due to CO_2 capture (%)	12.4
Equipment Availability (%)	100
Coal type	Shenhua bituminous coal
Plant operating life (years)	25
Discount rate (%)	10
CO_2 capture (g/kWh)	1127
CO_2 emissions rate (g/kWh)	125
CO_2 capture (%)	90
Compression pressure (bar)	110
Plant Thermal Input (GW)	1.4
Capital cost	
Base Plant total investment cost TIC (USD/kWh)	651
Base Plant TIC (million USD)	374
Capture Plant TIC (USD/$kW_{thermal}$)	122
Capture Plant TIC (million USD)	174
Overall Plant TIC (USD/kW_{net})	1376
Overall Plant TIC (million USD)	548
Project Contingency (10% of TIC) in million USD	54.8
Cost of Engineering & Design (7% of TIC) in million USD	38.4
Total Capital Cost (million USD)	641.2
Total Capital Cost (USD/kWnet)	***1610***
Operation and maintenance cost	
Coal Price ($/GJ) LHV basis	3.2
Total coal price (million USD/year)	104
Annual maintenance cost (4% of TIC) in million USD	22
CO_2 solvent & Consumable cost (million USD/year)	20
Tax and Insurance cost (2% of TIC) in million USD	3
Total Operation-maintenance Cost (million USD/year)	**149**
Levelized cost of electricity (USD/kWh)	0.07
Cost of CO_2 Abatement (USD/tCO_2)	***47.3***

temperature, 50 bar pressure and CO_2 stream (8) at mass flow rate of 377 tonne/hr., 106 °C temperature, 51 bar pressure are mixed in the mixture MIX1. After gaining heat from the heat exchanger B8 through heat integration, the mixed stream is transferred into the methanol reactor REACTR where CO_2 and H_2 is converted to methanol (MeOH), CO and H_2O along with unreacted H_2 and CO_2. Then at the outlet of the reactor, the stream (13) split into two streams through splitter B1 for the purpose of heat integration. The stream (S13) enters the separator B11, in which unreacted gaseous components and condensed liquid components are separated. The unreacted components are further passed through the second reactor B16 where further methanol yield is obtained. The products from the second methanol reactor B16 are again sent to separator B19 to separate the liquid and gaseous components. The liquid component from first separator B11 and second separator B19 are combined in the mixture B21 and then fed to third separator B12 to further remove liquid and gaseous components. The water and methanol are finally transformed to the distillation unit where they are separated from each other. Heat integration in the model is carried out through built-in energy analysis tool available in the Aspen Plus.

The methanol catalytic reactor model used in the flow sheet is a plug flow reactor that uses integral method to solve the mass, energy and species concentration equations along the reactor length for each differential element by accommodating detailed reaction kinetics of species. The plug flow reactor selected for the simulation model with constant inlet temperature type having an optimized length of 20 m and 0.1 m diameter achieved through sensitivity analysis. It is loaded with catalyst type $Cu/ZnO/Al_2O_3$ having particle density of 1775 kg/m^3 and loading of 150,000 kg. For the comparison purpose, same catalyst is adopted in the reactor bed for both Graaf and VB kinetic models. The distillation column is based on the equilibrium RadFrac model having 20 stages with partial-vapor type condenser and kettle type reboiler. The mixed stream (21) enters columns at stage 5 with operating specifications of distillate rate and reflux ratio are selected for the convergence of solution. To obtain the pure methanol with desired purity of 99%, a design specification block is created with model recovery as design parameter and reflux ratio as variable.

The thermodynamic properties of the gas phase species having pressure greater than 10 bar are computed using Redlich-Kwong-Soave MHV2 (RKSMHV2) equation of state while NRTL-RK model is used for pressure less than 10 bar. The pressure drop in the reactor bed is determined using Ergun's equation which is built-in model in the Aspen Plus. The reaction rate expressions form presented by Graaf cannot be directly implemented in Aspen Plus, rather reaction rate expressions need to be modified in specific types such as LHHW model to adjust in the built-in form, which is as follows:

$$\text{Rate of reaction} = \frac{[\textit{Kinetic factor}][\textit{Driving force}]}{[\textit{Adsorption term}]} = \frac{\left[kT^n e^{\left(-\frac{E_a}{RT}\right)}\right]\left[K_1\left(\prod C_i^{v_i}\right) - K_2\left(\prod C_j^{v_j}\right)\right]}{\left[\sum K_i\left(\prod C_j^{v_j}\right)\right]^m} \tag{6}$$

Where k = Pre exponent factor; n = Temperature exponent; T = Temperature; E_a = Activation energy; R = Universal gas law constant; C = Concentration component; m = adsorption expression exponent; K_1, K_2, K_3 = equilibrium constants; v = Concentration exponent; i, j = Component index.

The reaction rate expressions for the reactions (1–3) in Aspen built-in form used in the model as expressed by Kiss et al. [21], are as follows:

$$r_1 = \frac{\left[k_1 e^{\left(-\frac{E_{a,1}}{RT}\right)}\right]\left[K_{CO} f_{CO} f_{H_2}^{1.5} - \frac{K_{CO}}{K_1^{eq}} f_{MeOH} f_{H_2}^{-0.5}\right]}{[\textit{Adsorption term}]} \quad \text{(For reaction1)} \tag{7}$$

$$r_2 = \frac{\left[k_2 e^{\left(-\frac{E_{a,2}}{RT}\right)}\right]\left[K_{CO_2} f_{CO_2} f_{H_2} - \frac{K_{CO_2}}{K_2^{eq}} f_{H_2O} f_{CO}\right]}{[\textit{Adsorption term}]} \quad \text{(For reaction2)} \tag{8}$$

$$r_3 = \frac{\left[k_3 e^{\left(-\frac{E_{a,3}}{RT}\right)}\right]\left[K_{CO_2} f_{CO_2} f_{H_2}^{1.5} - \frac{K_{CO_2}}{K_3^{eq}} f_{H_2O} f_{MeOH} f_{H_2}^{-1.5}\right]}{[\textit{Adsorption term}]} \quad \text{(For reaction3)} \tag{9}$$

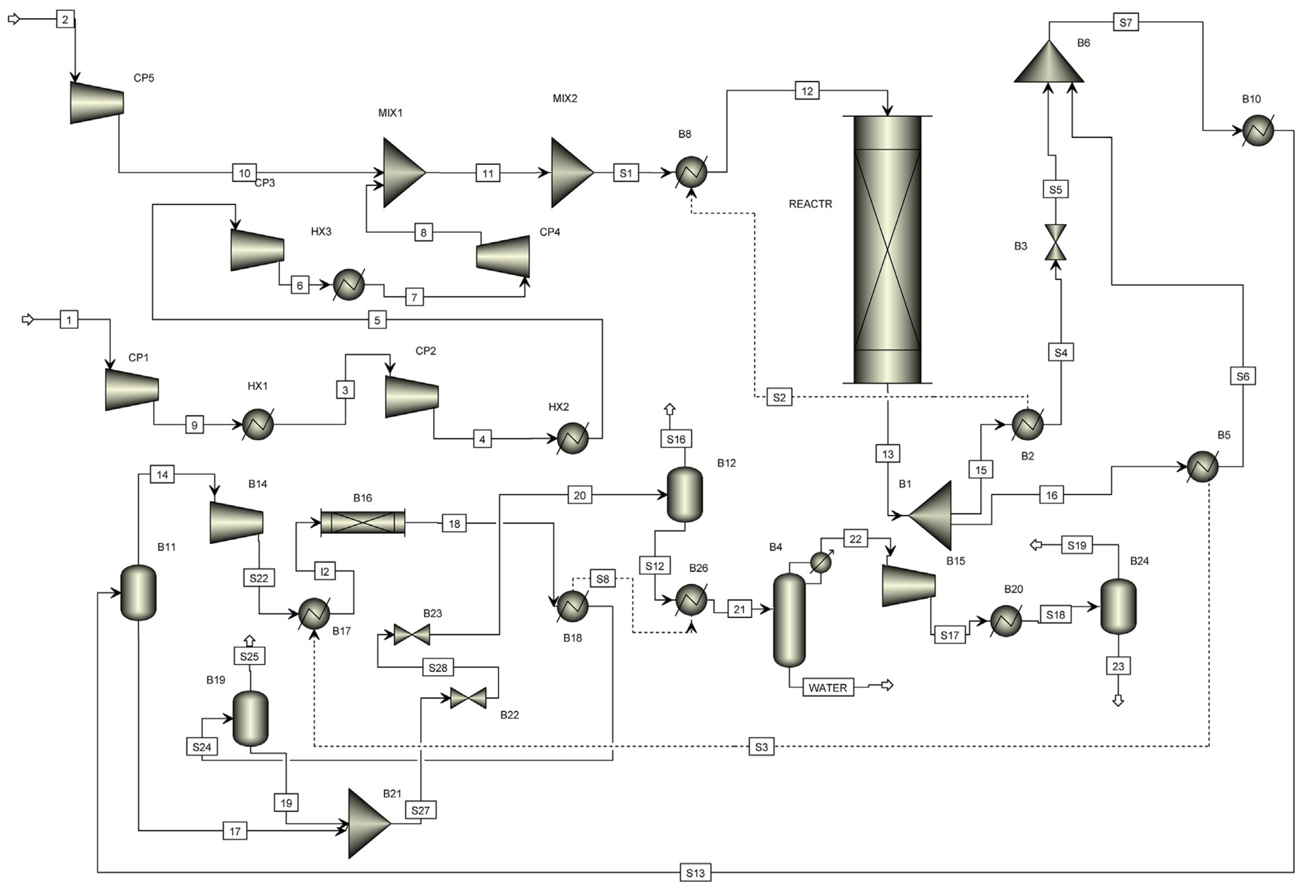

Fig. 2 – Flow sheet of methanol sysnthesis.

Where the adsorption term for all the three reactions (1–3) are same, and is expressed in Aspen Plus built-in form as follows:

$$\text{Adsorption term} = f_{H_2}^{0.5} + \frac{K_{H_2O}}{K_{H_2}^{0.5}} f_{H_2O} + K_{CO} f_{CO} f_{H_2}^{0.5} + \frac{K_{CO}K_{H_2O}}{K_{H_2}^{0.5}} f_{CO} f_{H_2O} + K_{CO_2} f_{CO_2} f_{H_2}^{0.5} + \frac{K_{CO_2}K_{H_2O}}{K_{H_2}^{0.5}} f_{CO_2} f_{H_2O} \tag{10}$$

In Aspen Plus setting, vapor is selected as reacting phase, concentration basis is partial pressure *(N/m^2)* instead of fugacity, and catalyst weight is selected for the rate basis with unit *kmol/kg-s* for the reaction rate. The parameters for the driving force in Aspen plus are in the following logarithmic form:

$$\ln(K) = A + \frac{B}{T} \tag{11}$$

The built-in adsorption expression used in Aspen Plus has the following form:

$$\text{Adsorption} = [Term1 + Term2 + \ldots]^q \tag{12}$$

Where $\{\ Term1 = K_i[C_1]^{n_1}[C_2]^{n_2}\ldots.\ \}$; q = adsorption expression exponent and for our study $q = 1$ as can also be observed from Eq. (10). Therefore the expression in Eq. (10) can be simplified in Aspen plus built-in form as follows:

$$\text{Adsorption term} = K_1 f_{H_2}^{0.5} + K_2 f_{H_2O} + K_3 f_{CO} f_{H_2}^{0.5} + K_4 f_{CO} f_{H_2O} + K_5 f_{CO_2} f_{H_2}^{0.5} + K_6 f_{CO_2} f_{H_2O} \tag{13}$$

where

$K_1 = 1; K_2 = \frac{K_{H_2O}}{K_{H_2}^{0.5}}; K_3 = K_{CO}; K_4 = \frac{K_{CO}K_{H_2O}}{K_{H_2}^{0.5}}; K_5 = K_{CO_2}; K_6 = \frac{K_{CO_2}K_{H_2O}}{K_{H_2}^{0.5}}$

Where the parameter K_i for the six terms is expressed in the following form:

$$K_i = e^A e^{\left(\frac{B}{T}\right)} T^c e^{(DT)} \tag{14}$$

The values for pre-exponent factor k, activation energy E_a, and temperature exponent n in Eq. (6), coefficients (A, B) for driving force for forward and reverse reaction rate in Eq. (11) and the values of coefficients (A and B) in Eq. (14) for the six terms in three reactions are taken from reference [60].

Likewise, to adjust the reaction rate expression presented by Vanden Bussche and Froment into Aspen Plus, some modifications are required, these reactions rate equations are modified according to reference [20] in Aspen Plus built-in form which are in the same general form as the Eq. (6)

$$\text{Rate of reaction} = \frac{[\textit{Kinetic factor}][\textit{Driving force}]}{[\textit{Adsorption term}]} = \frac{\left[kT^n e^{\left(-\frac{E_a}{RT}\right)}\right]\left[K_1\left(\prod C_i^{\upsilon_i}\right) - K_2\left(\prod C_j^{\upsilon_j}\right)\right]}{\left[\sum K_i\left(\prod C_j^{\upsilon_j}\right)\right]^m}$$

$$r_4 = \frac{\left[k_4 e^{\left(-\frac{E_{a,4}}{RT}\right)}\right]\left[K_1 P_{CO_2} P_{H_2} - K_6 P_{H_2O} P_{MeOH} P_{H_2}^{-2}\right]}{\left[1 + K_2 P_{H_2O} P_{H_2}^{-1} + K_3 P_{H_2}^{0.5} + K_4 P_{H_2O}\right]^3} \quad \text{(For reaction 4)} \tag{15}$$

$$r_5 = \frac{\left[k_5 e^{\left(-\frac{E_{a,5}}{RT}\right)}\right]\left[K_5 P_{CO_2} - K_7 P_{H_2O} P_{CO} P_{H_2}^{-1}\right]}{\left[1 + K_2 P_{H_2O} P_{H_2}^{-1} + K_3 P_{H_2}^{0.5} + K_4 P_{H_2O}\right]} \quad \text{(For reaction 5)} \tag{16}$$

The values of kinetic factor(k_4, k_5), activation energy($E_{a,4}$, $E_{a,5}$), driving force parameters (K_1, K_6, K_5, K_7) and adsorption parameters (K_2, K_3, K_4) are adopted from reference [61].

PEM electrolyzer system with single unit capacity of 2400 kg/day with total of 506 units, is used to produce total required H_2 of 52 tonne/hr. in stoichiometric proportion with CO_2. The levelised cost of H_2 from PEM electrolyzer is computed using the H2A hydrogen production tool v3.1 [62]. The total cost of hydrogen production consists of capital cost and operation maintenance cost. The total capital cost is further divided into direct capital and indirect capital cost. The detail of H_2 cost calculation is shown in Table 1. The capital cost of each unit is computed as scaled by the following equation [63]:

$$y = 224.49x^{0.6156} \tag{17}$$

where y: capital cost ($1000), x: unit capacity(kg/day)

Based on the above relation, the capital cost of hydrogen computed is 587.5 $/kW, which is further subdivide as 47% cost of stack and 53% cost of Bill of Plant (BOP). BOP includes cathode side H_2 management system, anode side O_2 management system, water supply system, heat control system, power electronics (transformer, rectifiers), and control system. The indirect capital cost includes cost of engineering (8% of direct capital cost), project contingency cost (15% of direct capital cost), construction, site preparation cost (2% of direct capital cost) and land cost.

Likewise, the operation and maintenance cost is also subdivided as fixed-operation maintenance cost and variable-operation maintenance cost. The Fixed-operation maintenance cost includes cost of annual planned maintenance (3% of direct capital cost), labor cost, and property tax and insurance cost (2% of total capital investment). Whereas, the variable-operation maintenance cost includes cost of electricity and cost of process water. To attain 98% availability of the equipment, a reasonable inventory of the equipment is essentially required which is regarded as replacement cost specially the stacks and crucial components of PEM electrolyzer are replaced after every seven years and replacement cost is taken as 15% of the installed capital cost. The estimated cost of H_2 production computed through the H2A tool is $3.9/kg of H_2 for the current required production rate and installed capacity.

The cost estimation of CO_2 abatement from 600 MW supercritical power plant located Guangxi China, is based on the conventional post-combustion CO_2 capture using 30 wt % MEA solvent. The detailed technical parameters of the plant integrated with CO_2 capture system and cost estimation is described in Table 2. The analysis is carried out by considering 90% CO_2 capture, 150 kW h/ton of CO_2 standard electricity energy input and compression of pure CO_2 stream to 110 bar. In order to calculate the cost of CO_2 abatement, the levelised cost of electricity is computed from the Chinese electricity tariff based on the coal generation power plant [64].

The major cost involved in the utilization of CO_2 to produce methanol is the cost of the input material stream such as H_2 and CO_2. The other costs involved are the capital cost that includes initial investment required for the equipment (methanol reactor, compressors, heat exchangers, distillation column, separator and condensers), cost for engineering and design, project contingency, cost of land, labor cost, maintenance cost, tax and insurance cost. The details of MeOH production cost is shown in Table 3. The capital cost estimation of the plant equipment is computed from the following equation as described in reference [8].

$$C = n^e C_0 \left(\frac{S}{nS_0}\right)^f \tag{18}$$

Where C: cost of the process unit; n: number of equally-sized units, C_0: cost of reference unit.

S: Capacity of the process unit; S_0: capacity of reference process unit; e: cost scaling exponent; f: cost scaling factor.

As the cost of process unit vary with the time and the equipment cost for methanol plant is available for the reference year 2008, therefore in order to compute present cost of the reference equipment, chemical engineering price cost index (CEPCI) is considered. The present cost of the reference unit is calculated from the following equation:

$$C_0' = C_0 \frac{C_i'}{Ci} \tag{19}$$

Where C_0: cost of reference unit for year 2008; C_0': cost of reference unit for year 2017.

C_i': Present value of CEPCI; Ci: Value of CEPCI for the reference year.

The value of CEPCI for the reference year (2008) is 575.4 while for present year (2017), it is 553 [65]. The cost of the compressor in capital cost calculation is neglected because the input stream such as CO_2 is supplied at pressure of 110 bar, and H_2 stream from PEM electrolyzer is obtained at pressure of 30 bar. Moreover, it is assume that no external thermal energy is required and heat integration within the system is sufficient to complete the methanol synthesis. Moreover, unreacted H_2 steam obtained at outlet of separator is utilized to meet required heat duty in the system. The annual capital cost is computed from the following relation by considering the initial investment, capital recovery factor, salvage value and discount rate:

$$\text{Annual capital cost} \approx (I - S) \times CRF + i \times S \tag{20}$$

Table 3 – Cost of methanol production.

Capital Cost	
Cost of 1st Methanol Reactor ($)	$ 47,049,047.60
Cost of 2nd methanol reactor ($)	$ 40,741,487.30
Cost of Methanol Distillation($)	$ 12,206,445.30
Cost of Compressor	Neglected
Cost of Heat Exchanger($)	$ 400,000.00
Total Initial investment($)	$ 100,396,980.20
Installation factor	1.12
Installed Capital Cost($)	$ 112,444,617.82
Cost of Engineering & Design ($): (8% of Total Initial Investment)	$ 8,031,758.42
Cost of Site Preparation ($): (2% of Total Initial Investment)	$ 2,007,939.60
Project Contingency ($): (15% of total Initial Investment)	$ 15,059,547.03
Cost of Land ($/acre)	$ 50,000.00
Land Required (acres)	10
Total Cost of Land ($)	$ 500,000.00
Salvage Value (5% of Total Initial Investment)	$ 5,019,849.01
Plant Life (Years)	25
Capital Recovery factor	0.084
Total Capital Cost ($)	**$ 138,043,862.87**
Annual capital Cost ($)	**$ 11,575,605.09**
Fixed Operating Cost	
Plant Operators	10
Yearly salaries and overheads ($)	$ 50,000.00
Total Labor Cost ($)	$ 500,000.00
Maintenance and Repairs cost ($/year): 3% of Installed Capital Cost	$ 3,373,338.5
Tax and Insurance (2% of Annual Capital Investment)	$ 231,512.1
Annual Fixed Cost	**$ 4,104,850.6**
Variable Operating Cost	
Cost of Catalyst ($/kg)	106.6
Total Catalyst Consumption (kg)	180,000
Total Cost of Catalyst ($)	$ 19,188,000.00
Cost of H_2 ($/kg)	3.9
H_2 Consumption (kg/Year)	442,975,680
Total Cost of H_2 ($/year)	$ 1,727,605,152.00
Cost of CO_2 abatement $/tonne CO_2	47.3
Total CO_2 Utilized (tonne/year)	3236470
Total Cost Of CO_2 ($/year)	$ 153,085,031.00
Total Variable Cost ($/year)	**$ 1,899,878,183.00**
Total Cost ($/Year)	**$ 1,915,558,638.7**

where. I: Initail invesment;S: Salvage value; CRF: Capacity recovery factor;i: Discount rate

The capital recovery factor is as follows:

$$CRF = \frac{i\,(1+i)^N}{(1+i)^N - 1} \tag{21}$$

Where. N : Plant life

Results and discussion

Methanol production is analyzed based on the two kinetic models such as Graaf model and VB model. At first, the simulation models developed in the Aspen Plus are validated with the experimental data available in the literature. As shown in Fig. 3a, the predicted mole fraction of species obtained from the Graaf model is compared with the experimental data of reference [60] at various reactor inlet temperatures and shows that the simulation results are in good agreement with the experimental data. Likewise VB model based mole fraction variation along the reactor length is also in good agreement with the results of reference [12] as shown in Fig. 3b. After validation, the crucial parameters such as catalyst loading, reactor operating pressure and temperature are optimized through sensitivity analysis of the parameters.

As shown in Fig. 4 the reactor inlet temperature is varied from 150 °C to 300 °C to predict the optimal operating temperature of methanol reactor. The results indicate that optimal inlet reactor temperature based on VB model is 240 °C at which 16.6% yield of methanol is obtained, whereas for Graaf model, the optimal temperature is 210 °C at which 26.2% yield of methanol is achieved. It is clear from the results that by changing the kinetic reactor model, the optimal yield obtained by Graaf model is about 9.6 point percentage higher than the VB model.

Similarly, reactor pressure is varied from 10 to 80 bar and results in Fig. 5 for both VB and Graaf model indicate that methanol yield steadily increases by increasing the reactor operating pressure. However, 50 bar reactor operating pressure is selected for the present study because of the limitation of the validity range of the kinetic models. The impact of catalyst loading on the methanol yield is shown in Fig. 6. The catalyst loading is varied from 4000 kg to 200,000 kg in the reactor bed. The results depicted that the increase in methanol yield is prominent up to 150,000 kg bed of catalyst, beyond which trivial impact is observed on methanol yield.

The suitable stoichiometric ratio between CO_2 and H_2 influence the performance of the reactor. The impact of ratio between mass flow rate of CO_2 and H_2 (CO_2/H_2) on methanol is shown in Fig. 7. The results indicated that when CO_2/H_2 mass flow ratio was varied from 6 to 88, methanol yield was decreased from 18.4% to 0.6% for VB model and 29.9%–1.4% for Graaf model. After predicting optimal conditions of the methanol reactor, complete methanol production plant is operated to measure yield of the plant using two types of the kinetic reactor model. In distillation column, design specifications are used to achieve mole recovery of 99% methanol, while reflux ratio is chosen as adjusted variable. For Graaf model, the predicted methanol production is 119 tonne/hr. and CO_2 abated is 179.9 tonne/hr. On the other hand, for VB model the methanol production is 99 tonne/hr with 164.5 tonne/hr. of CO_2 abatement. Hence, Graaf model predicted 20 tonne/hr of methanol and 15.4 tonne/hr of CO_2 abatement more than the VB model for the same input. It means that Graaf model predicted 0.66 tonne of methanol per tonne of mitigated CO_2 which is 9% higher than the methanol yield predicted by VB model which is 0.6 tonne of methanol per tonne of CO_2 mitigated. This indicates that the selection of kinetic model changes the output yield of the plant, hence cost calculations based on each kinetic model can produce different estimation. The mass balance of the important streams used in methanol synthesis plant, molar flow, temperature and pressure is shown in Table 4.

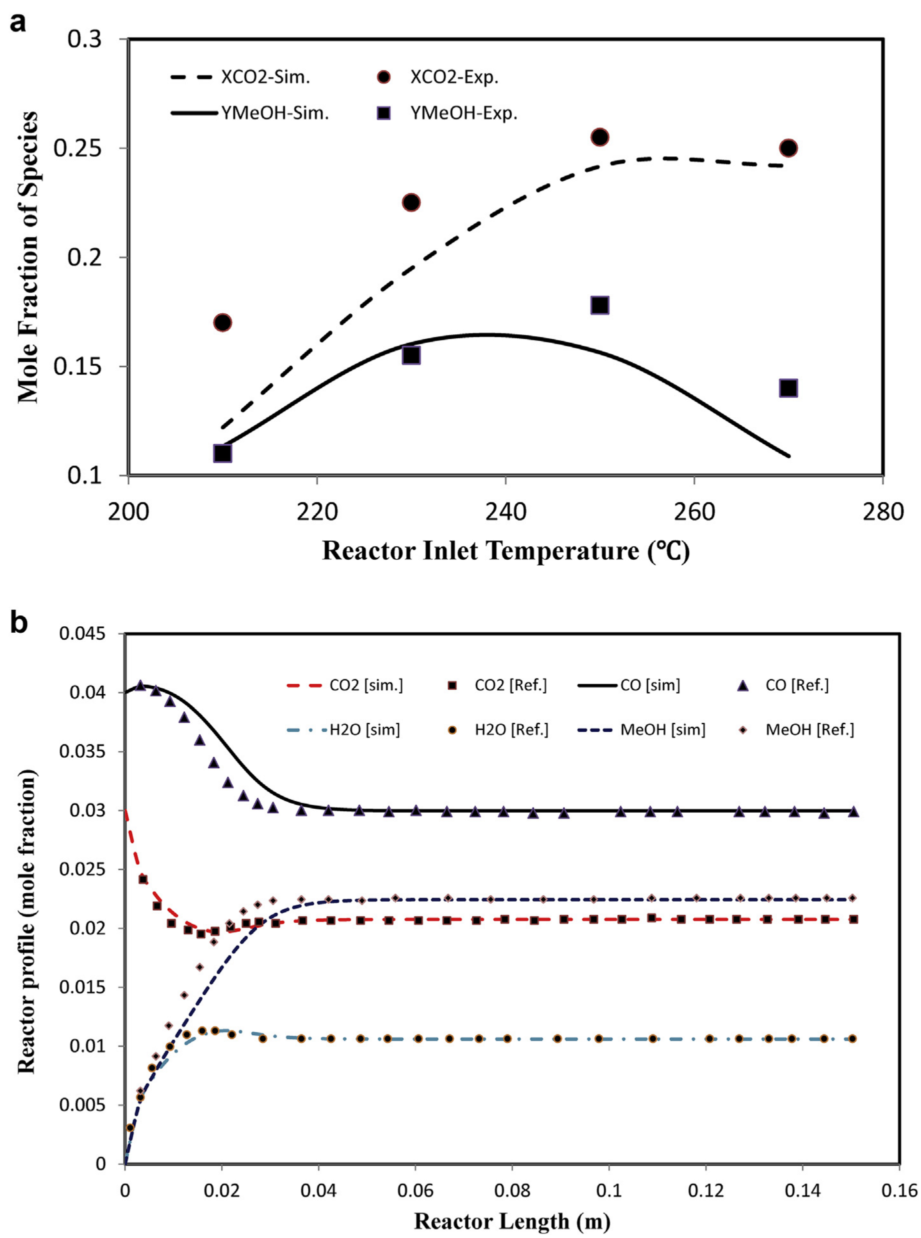

Fig. 3 – (a-b): Comparison of simulation results with experimental data of methanol reactor. (a) For Graaf model, XCO2: CO_2 conversion, YMeOH: methanol yield. (b) For VB model.

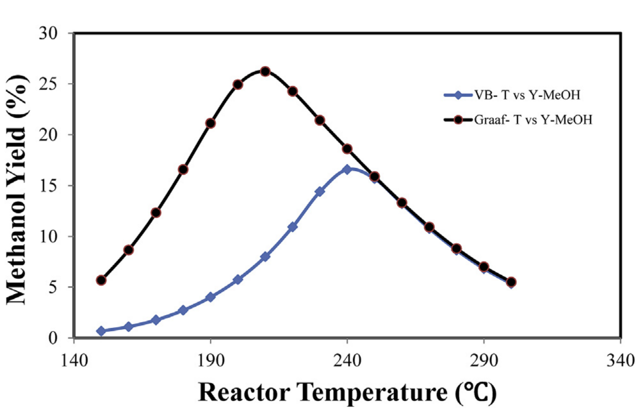

Fig. 4 – Methanol yield at various inlet temperature of methanol reactor.

Economic analysis

The methanol production plant is integrated with hydrogen generation and CO_2 captured from coal power plant, whereas the hydrogen generation is coupled with the large scale hydropower plant. To compute the overall levelised cost of methanol production system, the cost involved in the hydrogen production and cost of CO_2 mitigation is calculated separately. Hydrogen generation cost is computed based on the input electricity and water feed from hydropower plant. China has installed world largest hydro power plant, and cost of electricity generation through such a large dam is low. The Three Gorges dam installed in China has total capacity of 22.4 GW with 32 units where each has capacity of 700 MW. The estimated levelised cost of electricity generation through this

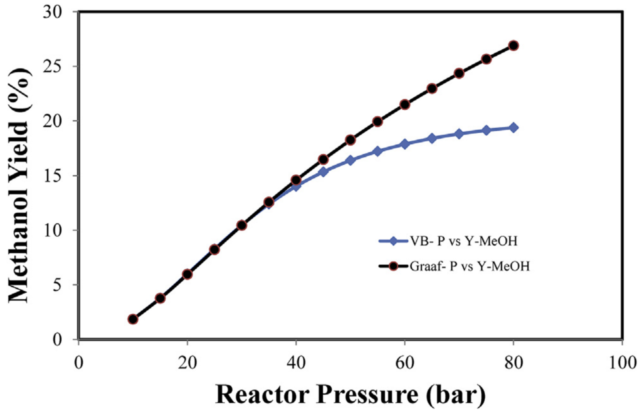

Fig. 5 – Methanol yield at various inlet pressure of methanol reactor.

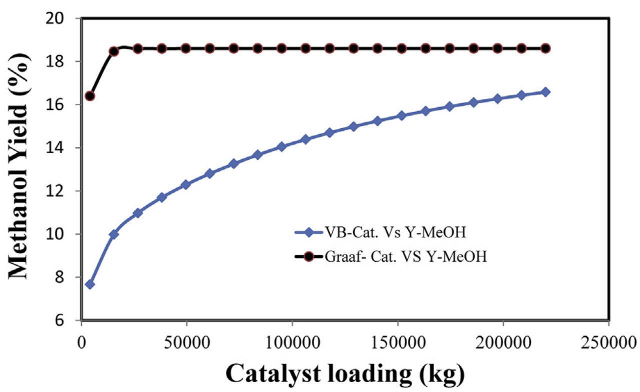

Fig. 6 – Impact of catalyst loading on methanol yield.

large scale dam is USD 0.016/kWh. This cost is used to further do the cost analysis of hydrogen generation, and methanol synthesis. The technical and economic performance of CO_2 mitigation from 600 MW supercritical coal power plant located at Guangxi China is based on post-combustion CO_2 capture using conventional absorber-stripper arrangement and 30 wt % MEA as chemical solvent. Based on the China's local circumstances, for 90% CO_2 capture, 110 bar compressed CO_2 and

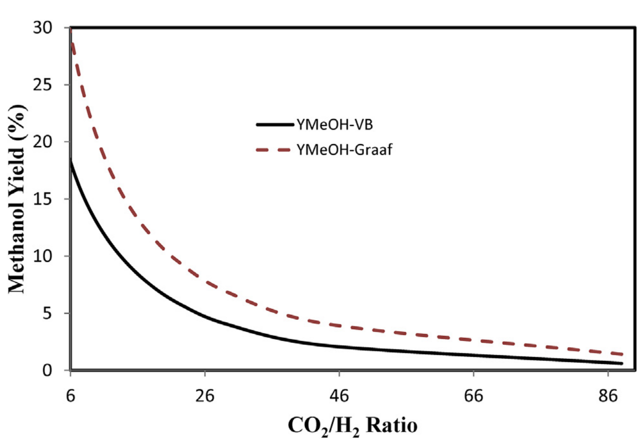

Fig. 7 – Impact of CO_2/H_2 ratio on methanol yield.

Table 4 – Mass balance of the streams used in methanol production plant.

Stream	1	3	4	6	7	9	11	12	13	14	15	16	17	19	20	22	23
Mole Flow (kmol/sec)																	
CO_2	2.38	2.38	2.38	2.38	2.38	2.38	2.38	2.38	1.71	1.64	1.02	0.68	0.06	0.05	0.11	0.01	0.01
CO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.06	0.06	0.04	0.02	0.00	0.00	0.00	0.00	0.00
H_2O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.01	0.40	0.27	0.67	0.46	1.13	0.01	0.01
CH3OH	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.62	0.03	0.37	0.25	0.59	0.47	1.06	1.03	1.03
H_2	0.00	0.00	0.00	0.00	0.00	0.00	7.17	7.17	5.26	5.24	3.16	2.10	0.02	0.01	0.03	0.00	0.00
Mole Fraction																	
CO_2	1.00	1.00	1.00	1.00	1.00	1.00	0.25	0.25	0.21	0.24	0.21	0.21	0.05	0.05	0.05	0.01	0.01
CO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.00	0.00	0.00	0.00	0.00
H_2O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.00	0.08	0.08	0.50	0.46	0.48	0.00	0.00
CH_3OH	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.07	0.00	0.07	0.07	0.44	0.48	0.46	0.98	0.99
H_2	0.00	0.00	0.00	0.00	0.00	0.00	0.75	0.75	0.63	0.75	0.63	0.63	0.01	0.01	0.01	0.00	0.00
Mass Flow tonne/hr.																	
CO_2	377.00	377.00	377.00	377.00	377.00	377.00	377.00	377.00	270.17	260.41	162.10	108.07	9.76	7.32	17.08	1.99	0.84
CO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.95	5.93	3.57	2.38	0.02	0.02	0.04	0.00	0.00
H_2O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43.73	0.41	26.24	17.49	43.32	29.69	73.01	0.33	0.33
CH_3OH	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	70.97	3.30	42.58	28.39	67.67	54.72	122.39	119.06	118.59
H_2	0.00	0.00	0.00	0.00	0.00	0.00	52.00	52.00	38.18	38.06	22.91	15.27	0.12	0.09	0.21	0.00	0.00
Temperature (°C)	25.00	38.00	157.92	151.02	38.00	132.80	92.22	210.00	210.00	35.00	210.00	210.00	35.00	35.00	25.61	64.08	40.00
Pressure (bar)	1.00	2.70	8.80	26.30	26.00	3.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	1.20	1.00	1.00

at 27.9% net efficiency, the levelised cost of CO_2mitigation is computed to be 47.3 $/ton CO_2.

Proton exchange membrane (PEM) electrolysis technique is adopted to compute the cost of hydrogen production. The total cost of hydrogen generation is composed of the direct capital cost, indirect capital cost, fixed and variable operating cost, and replacement cost. As shown in Fig. 8 the major contribution in the cost of H_2 generation involves the cost of electricity, initial capital investment and fixed operating cost. For the current study, it is assumed that the electricity is provided through renewable hydropower plant of large capacity such as The Three Gorges in China, in which cost of electricity is low and abundant water is also available for electrolysis. Based on the required capacity of 52 tonne/hr., 98% capacity factor, and 40 years of plant life, the levelised cost of H_2 production is 3.9 USD/kg H_2.

The cost of methanol synthesis by catalytic hydrogenation of CO_2 consists of capital cost, fixed and variable cost. The cost breakdown of the total yearly cost involved for the production of methanol is shown in Fig. 9 which indicates that 90.2% of the total cost is the cost of hydrogen, 8% is the cost of CO_2 abatement, 0.6% is the total capital cost and 0.2% is the total fixed operating cost.

The total value of methanol produced annually through CO_2 hydrogenation and oxygen as a byproduct from the PEM electrolysis is shown in Table 5. The current market price of methanol is obtained from reference [66], whereas cost of oxygen is taken from reference [67]. The results indicate that the net value of the methanol produced annually is about 0.51 billion USD while the net value of the oxygen produced annually as a byproduct in the PEM electrolysis is about 0.22 billion USD. Therefore total worth of the products produced annually in the system is about 0.73 billion USD, whereas total annual expenditures on the production of the methanol is about 1.9 billion USD. Therefore, to mitigate 180 tonne/hr. of CO_2 to produce methanol, an additional cost of 1.2 billion USD annually has to be consumed with the current cost of H_2 generation through PEM electrolysis and with current technology for CO_2 capture.

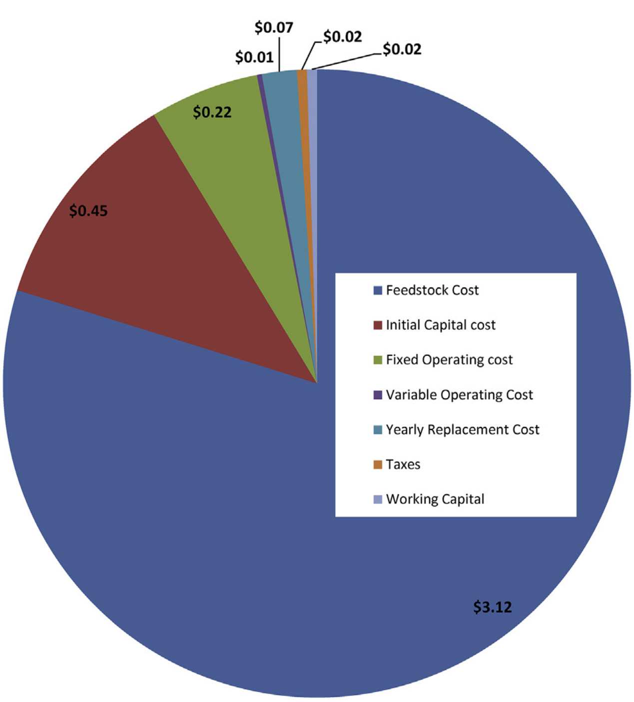

Fig. 8 – Breakdown of H_2 production cost.

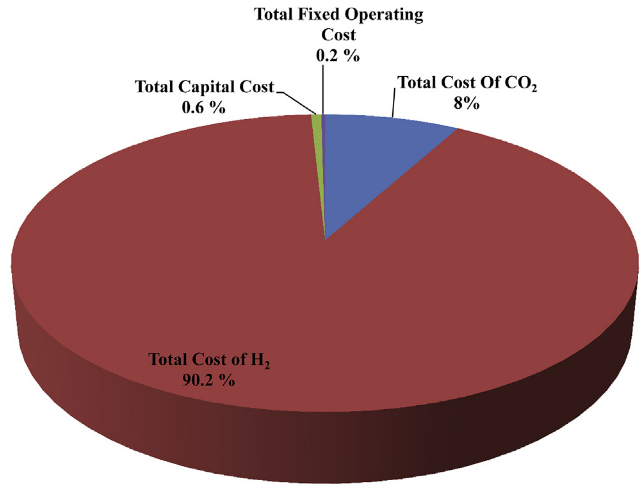

Fig. 9 – Breakdown of methanol production cost.

The impact of crucial factors such as production rate of methanol, cost of hydrogen generation, cost of variable expenses, market sale price of methanol and oxygen, on the total worth of products produced and on total cost of production is analyzed through sensitivity analysis. The cost breakdown of methanol production indicates that 90.2% of the methanol production cost is related to hydrogen production, and 8% of total cost is due to CO_2 mitigation.

The impact of important economic factors such as H_2 production cost, CO_2 mitigation cost and variable cost on total cost of methanol production and on total revenue generated is shown in Fig. 10. The levelised cost of hydrogen generation and levelised cost of CO_2 mitigation is taken as 40%, 50%, 75% and 90% of the base value. 10. As in our base case calculation the maximum yield of the methanol is 26.2%. However due to recent advances in catalyst development, the CO_2 conversion to methanol is expected to increase in the near future. To analyze the impact of methanol production rate, total cost of production is compared with three scenarios such as with base case, 1.5 times increase in production and with 2 times increase in production rate as indicated in Fig. 10. The results indicate that for the base case production rate, methanol production is not feasible even if the cost of hydrogen production is taken as 40% of the base case or variable cost is

Table 5 – Worth of methanol and oxygen produced annually (USD).

Total Methanol Produced (tonne/year)	1021591.2
Current Price of Methanol ($/tonne)	504
Worth of Methanol Produced($/year)	$ 514,881,964.80
Total Oxygen Produced (tonne/year)	3543805.44
Price of Oxygen ($/tonne)	60.7
Worth of Oxygen Produced ($/year)	$ 215,108,990.21
Total value of the products	$ 729,990,955.01

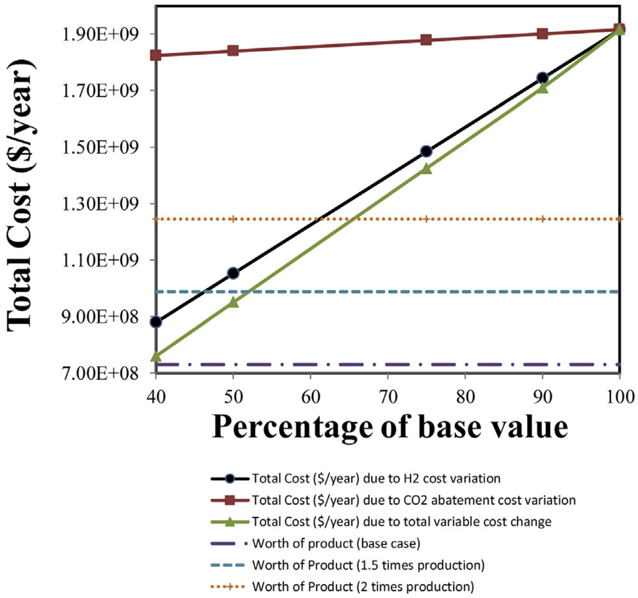

Fig. 10 – Impact of H_2, O_2 and variable cost reduction.

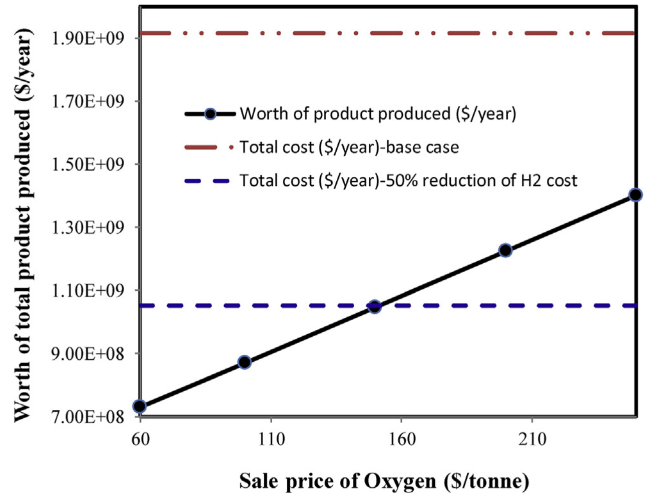

Fig. 12 – Impact of increase in oxygen selling price.

taken as 40% of the base case. However, process becomes economically feasible if the methanol production rate is enhanced to 1.5 times of the base value and hydrogen cost is taken as 46% of the base value or variable cost is taken as 52% of the base value.

Likewise, market selling price of methanol and oxygen vary with the demand and it may further increase in the near future because of increasing use of methanol in various chemical manufacturing and as a fuel. To analyze the impact of methanol and oxygen selling price on total cost and worth of the product, the methanol selling price is increased from 500 $/tonne to 1000 $/tonne (Fig. 11), whereas oxygen selling price is increased from 60 $/tonne to 250 $/tonne as shown in Fig. 12. The results of Fig. 11 indicate that if methanol selling price is increased from current market price (500 $/tonne) to 1000 $/tonne, the methanol production for the base case hydrogen production price will not be feasible. However, if the hydrogen production price is reduced by 50% and methanol selling price is set at 820 $/tonne, then methanol process become profitable. The impact of increase in oxygen selling price indicates that for the base case hydrogen price, methanol production will not be feasible even if the oxygen selling price is increased from 60 $/tonne to 250 $/tonne. However, by reducing hydrogen price to 50%, and by selling oxygen at price of 150 $/tonne, the methanol production becomes feasible. As mention in the reference articles [22] and [23] that with current cost of hydrogen and electricity, methanol synthesis is not economically feasible. However, sensitivity analysis indicates that methanol production becomes economically feasible in three ways: first option is if the methanol production rate is enhanced to 1.5 time of the base value and hydrogen cost is taken as 46% of the base value or variable cost is taken as 52% of the base value. The second way if the hydrogen production price is reduced by 50% and methanol selling price is set at 820 $/tonne, third way is by reducing hydrogen price to 50% and by selling oxygen at price of 150 $/tonne.

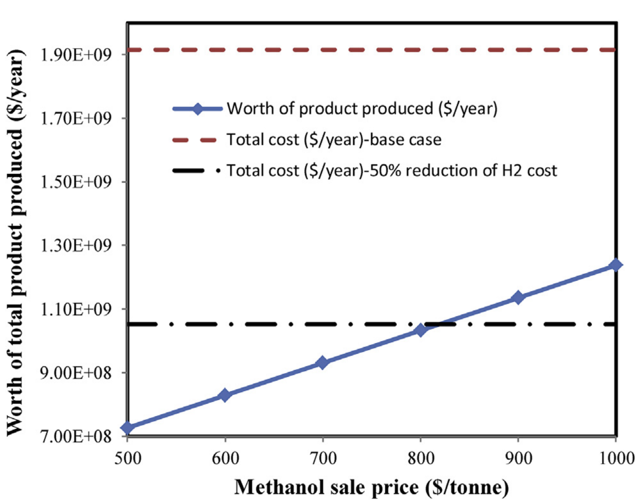

Fig. 11 – Impact of increase in methanol selling price.

Conclusion

This work has technically and economically analyzed the hydrogen generation through PEM electrolysis by adopting electricity and water from large scale hydropower plant and CO_2 capture from 600 MW supercritical coal power plant integrated with direct methanol production plant. At first, the catalytic reactor for CO_2 hydrogenation to methanol is modeled and simulated to predict the reactor optimal conditions based on two different kinetic reaction models such as Graaf kinetic model and VB kinetic model. Then the optimized reactor model is extended to model complete flow sheet of direct methanol production plant. Results indicated that for the same input, Graaf model produced 0.66 tonne of methanol per tonne of CO_2 mitigated which is 9% higher than VB model that predicted 0.6 tonne of methanol per tonne of CO_2.

The economic feasibility of methanol production by utilizing captured CO_2 from coal power plant and hydrogen generation from PEM electrolysis is also carried out in this study. The water and electricity required for the electrolysis is taken from the large hydropower plant. By comparing the annual worth of methanol production and oxygen generation in the electrolyzer as byproduct with the annual cost of production, it is evident that with current price of hydrogen and CO_2 mitigation cost, methanol production is not cost effective because to mitigate 180 tonne/hr. of CO_2 to produce methanol, an extra cost of 1.2 billion USD annually is required. As mention in the reference articles [22] and [23] that with current cost of hydrogen and electricity, methanol synthesis is not economically feasible. However, sensitivity analysis indicates that methanol production becomes economically feasible in three ways: first option is if the methanol production rate is enhanced to 1.5 time of the base value and hydrogen cost is taken as 46% of the base value or variable cost is taken as 52% of the base value. The second way if the hydrogen production price is reduced by 50% and methanol selling price is set at 820 $/tonne, third way is by reducing hydrogen price to 50% and by selling oxygen at price of 150 $/tonne.

In order to make catalytic CO_2 hydrogenation economically feasible, three recommendations are proposed. Firstly, the CO_2 conversion to methanol is about 26% in the reactor with the commercially available catalyst, this CO_2 conversion can be increased up to 50% by improving the catalytic efficiency and selectivity of the catalyst material. Secondly, CO_2 mitigation cost considered in the current study is based on the conventional MEA solvent which requires huge energy to regenerate solvent. Some new solvents such as ammonia based solvents and ionic liquids have shown higher capture efficiency at expense of low heat duty for solvent regeneration. Thirdly, cost of hydrogenation in electrolysis is main influential factor in methanol synthesis. This cost can be decreased by reducing the electricity consumption per kg of H_2 produced ($kWh/kg H_2) and can be achieved by improving the stack design, devising low cost and efficient electrode material to enhance the Faradaic and energy efficiency of the electrolyzer.

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